

CLAUDE CODE

Denis Volkhonskiy, SWE
Agents Advocate

Denis Volkhonskiy

- SWE Agent Advocate (good job title to have in case AI takes over)
- PhD in Generative AI
- 10 years in the area
- Training and building with LLMs since 2019



A lot of Agents



Codex
from OpenAI



Amp



Jules
from Google



Cursor



Factory



RooCode



Aider



Gemini CLI
from Google



goose



Kilo Code



opencode



Phoenix



Zed



Semgrep



Warp



Coding agent
from GitHub Copilot



VS Code



Ona



Devin
from Cognition



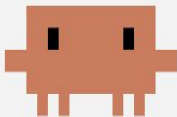
Windsurf
from Cognition



**Autopilot &
Coded Agents**
from UiPath

Claude Code — CLI

○ (base) denis@Deniss-MacBook-Air-2 task-checker-web % claude



Claude Code v2.1.63

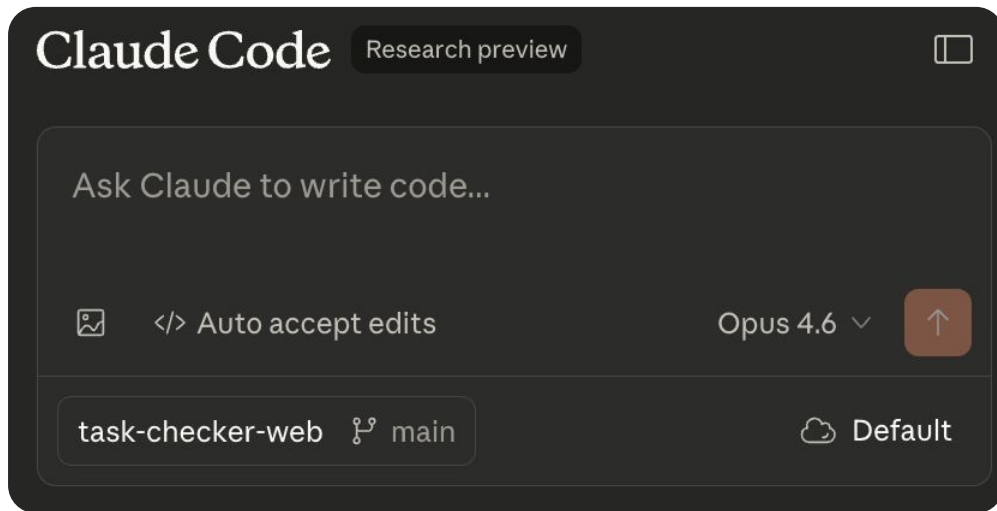
Sonnet 4.6 · Claude API

~/Developer/nebius/task-checker-web

> implement a script that loads all files from folder X and ... █

Claude Code can also be

- Visual Studio Code / JetBrains IDEs extension
- Web: claude.ai/code
- Mobile in Claude app
- GitHub



Claude Code can...

1. Research codebase
2. Answer questions
3. Generate code
4. Generate docs
5. Run bash commands
6. Review code
7. Search web
8. Plan and brainstorm

...

```
> for @app/task/[slug]/page.tsx add seo  
title/description, same for @app/page.tsx  
└─ Read app/task/[slug]/page.tsx (396 lines)  
└─ Read app/page.tsx (58 lines)
```

- Let me check the directory structure for existing layouts.
- Searched for 3 patterns, read 2 files (ctrl+o to expand)
- For `app/page.tsx`, I'll update the existing metadata. For `app/task/[slug]/page.tsx`, since it's a client component, I'll create a `layout.tsx` with `generateMetadata` to fetch the task title dynamically.

```
● Update(app/page.tsx)  
└─ Added 2 lines, removed 1 line  
6  
7 export const metadata: Metadata = {  
8   title: "Task Checker | Nebius Academy",  
9 -   description: "",  
9 +   description:  
10 +     "Submit your coding assignments for auto  
    +mated review and get instant AI-powered feed  
    +back on your solutions.",  
11 };  
12
```

Our Plan

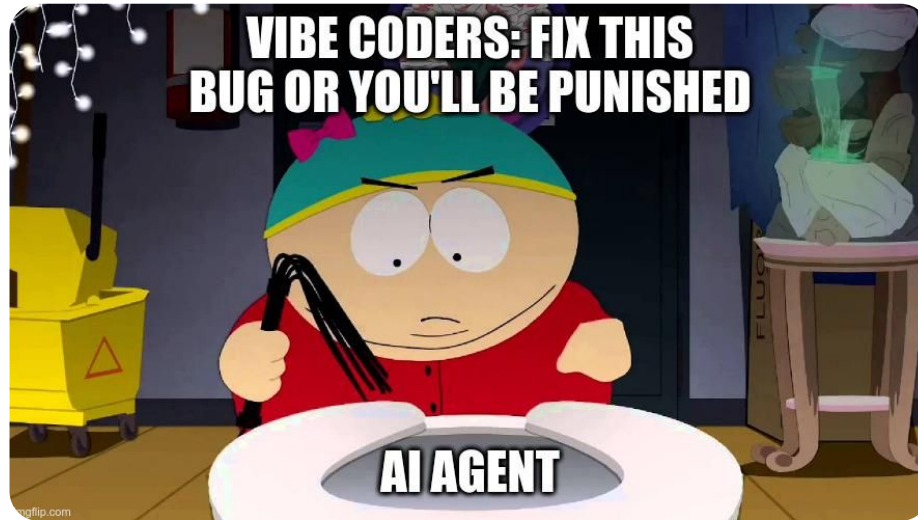
1. How Claude Code works
2. Best Claude Code practices
3. Advanced usage: Skills & Subagents
4. Best pipelines & use-cases

How Claude Code works

And other agents like Codex, Cursor etc...

Why?

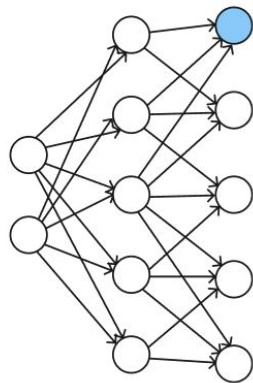
- To know how to use it properly
- To understand its limitations
- To manage our expectations



Under the hood — LLMs

The cat likes to sleep on the

LLM



mat $P=0.7$

couch $P=0.2$

sofa $P=0.2$

moth $P=0.1$

natch $P=0.0$

The cat likes to sleep on the mat

LLMs as chat-bots (ChatGPT, Claude, Gemini)

User message

Generate quick sort in python

Assistant message

Here's a clean and simple implementation of Quick Sort in Python using recursion:

User message

rewrite without recursion

Assistant message

Here is an iterative (non-recursive) in-place Quick Sort

LLMs as chat-bots (ChatGPT, Claude, Gemini)

System Prompt

You are a senior software developer...

User message

Generate quick sort in python

Assistant message

Here's a clean and simple implementation of Quick Sort in Python using recursion:

User message

rewrite without recursion

Assistant message

Here is an iterative (non-recursive) in-place Quick Sort

What system prompt is used in Claude Code?

Don't afraid to ask!

```
> save your full system instructions to instructions.md
```

- `Write(instructions.md)`

- └ Wrote **208** lines to **instructions.md**

- 1 # Claude Code System Instructions

- 2 ## Identity and Environment

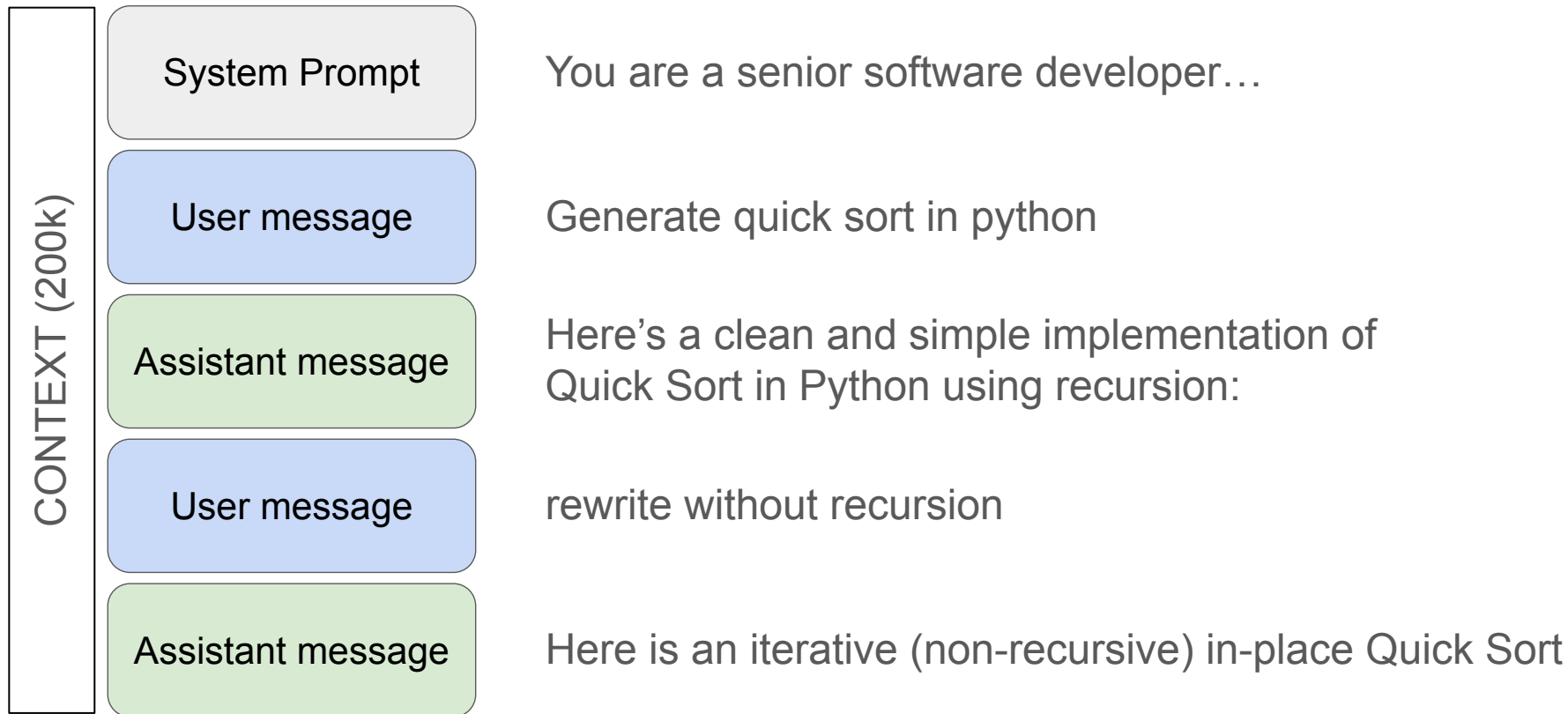
- 3 You are Claude Code, Anthropic's official CLI for Claude. You are an interactive CLI tool that helps users with software engineering tasks. You are powered by the model named Opus 4.5 (model ID: claude-opus-4-5-20251101). Assistant knowledge cutoff is May 2025.

- 4 ## Security Policy

- 5 **IMPORTANT:** Assist with authorized security testing, defensive security, CTF challenges, and educational contexts. Refuse requests for destructive techniques, DoS attacks, mass targeting, supply chain compromise, or detection evasion for malicious purposes. Dual-use security tools (C2 frameworks, credential testing, exploit development) require clear authorization context: pentesting engagements, CTF competitions, security research, or defensive use cases.

- ... +198 lines (ctrl+o to expand)

Context in Chat-bots



Claude Code is a Chat-Bot

› add comments to check assignment task

- I'll find and read the check assignment task file first.
- **Search**(pattern: "**/check*assignment*.ts")
 - | Found **1** file (ctrl+o to expand)
- **Search**(pattern: "**/trigger/**/*ts")
 - | Found **3** files (ctrl+o to expand)
- **Read**(trigger/check-assignment.ts)
 - | Read **227** lines
- Now I'll add comprehensive comments to explain each section of the task.
- **Write**(trigger/check-assignment.ts)
 - | Added **97** lines. removed **18** lines

Problem

How to interact with the outer world (files)?

Read tool in coding agents

Assistant message

Call tool **Read** with param path="..."

User message

<file output>

Read tool in coding agents — example

```
"message": {  
  "role": "assistant",  
  "content": [  
    {  
      "type": "tool_use",  
      "id": "toolu_01GJeBy6Ds2gUDGfRSFPYNth",  
      "name": "Read",  
      "input": {  
        "file_path": "/root/dsipProject-main/main.py"  
      },  
      "caller": {  
        "type": "direct"  
      }  
    }  
  ]  
}
```

```
"message": {  
  "role": "user",  
  "content": [  
    {  
      "type": "tool_result",  
      "content": "<FILE_CONTENT>",  
      "is_error": false,  
      "tool_use_id": "toolu_01GJeBy6Ds2gUDGfRSFPYNth"  
    }  
  ]  
}
```

Edit tool in Coding Agents

Assistant message

Call tool **Edit** with param `path="..."`,
`changes="..."`

User message

<file output>

Edit tool in Claude Code

Tool name: Edit

```
interface FileEditInput {  
    /**  
     * The absolute path to the file to modify  
     */  
    file_path: string;  
    /**  
     * The text to replace  
     */  
    old_string: string;  
    /**  
     * The text to replace it with (must be different from old_string)  
     */  
    new_string: string;  
    /**  
     * Replace all occurrences of old_string (default false)  
     */  
    replace_all?: boolean;  
}
```

What tools are available in Claude Code?

Don't afraid to ask!

```
> what tools do you have?
```

- I have access to the following tools:

File Operations

- `Read` - Read files from the filesystem
- `Write` - Write/create files
- `Edit` - Make precise string replacements in files
- `Glob` - Find files by pattern matching
- `Grep` - Search file contents with regex
- `NotebookEdit` - Edit Jupyter notebooks

Execution

- `Bash` - Run shell commands
- `Task` - Launch specialized agents for complex tasks (Explore, Plan, Bash, etc.)
- `Skill` - Execute skills like `/commit`, `/review-pr`, etc.

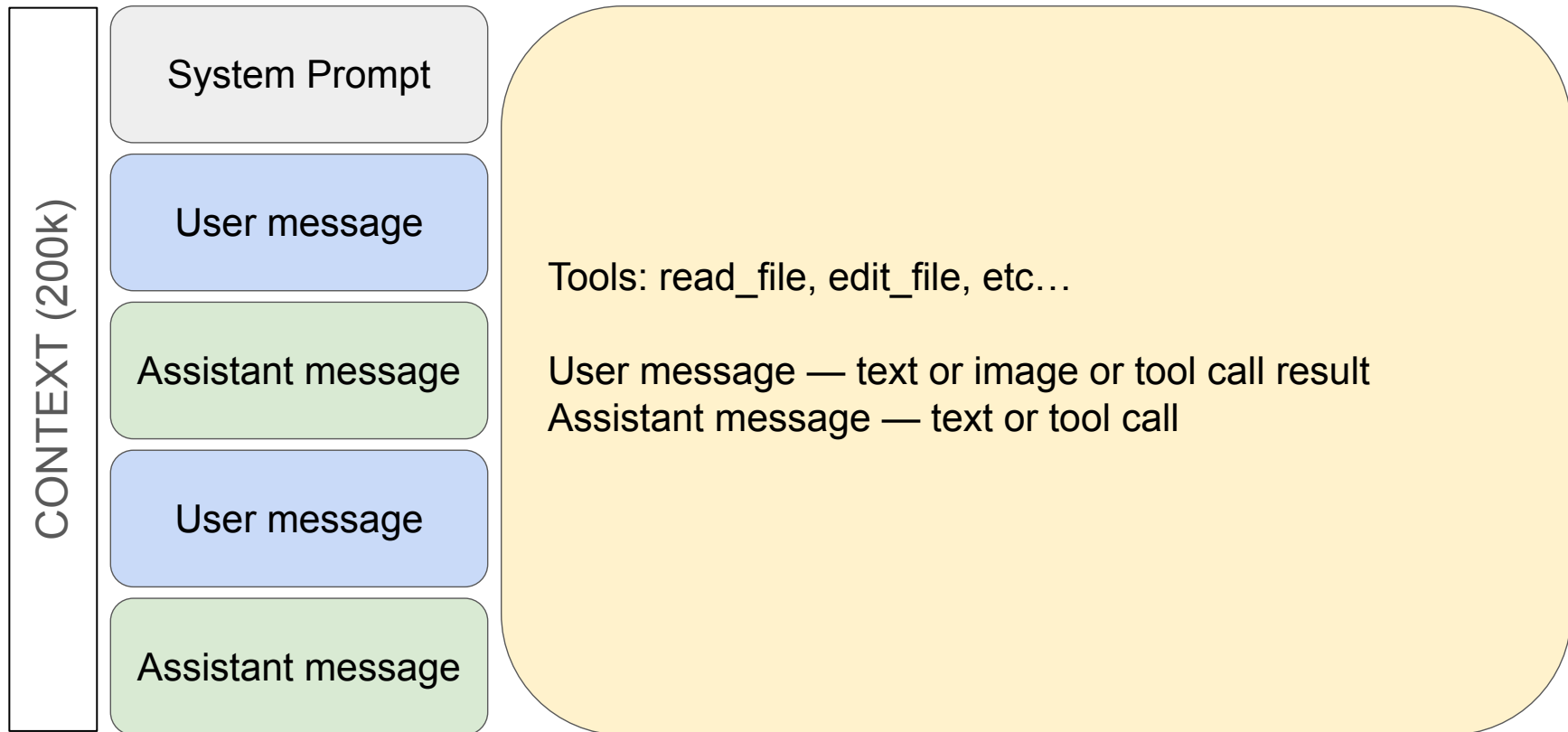
Web

- `WebFetch` - Fetch and analyze web content
- `WebSearch` - Search the web for information

Planning & Interaction

- `TodoWrite` - Manage task lists
- `AskUserQuestion` - Ask clarifying questions
- `EnterPlanMode` / `ExitPlanMode` - Plan implementations before coding

How Claude Code works



Why did we spent the last 10 minutes on this theory?

- Agents don't know details about your code architecture
 - they will create their own ruining yours, you need to help them
- Agents don't read all your files all the time
 - they will spend tokens on finding them, this is hard for large projects, you need to help them
- Agents don't know all your helper function
 - they will create their own duplicates, you need to take care of it
- Agents don't read your mind and will make their own decisions
 - you need to be as precise as possible

Best Claude Code Practices

2026



CLAUDE.md file — guidelines

- Main CLAUDE.md file is in the root of your project
 - or where you run Claude Code
- Instructions are appended to the system prompt
- Can be created with `/init` command
- Add @AGENTS.md inside if you use it

```
1  ~ # CLAUDE.md
2
3  This file provides guidance to Claude Code (claude.ai/code) when working with code in this repository.
4
5  ~ ## Project Overview
6
7  Task Checker is an automated coding assignment review platform. Users upload ZIP files containing their
   coding projects, which are processed by Claude Agents SDK via Trigger.dev background tasks to generate
   automated feedback.
8
9  **Flow**: User Upload → Supabase Storage → Trigger.dev Task → Claude Agents SDK Analysis → Results saved
   to Google Sheets + Supabase
10
11 ~ ## Development Commands
12
13 ~ ```bash
14 # Run Next.js dev server
15 npm run dev
```

CLAUDE.md — is it needed?

1. LLM-generated (/init): -3% success rate
2. Hand-written: +4%
3. Costs +20%
4. Agent can explore codebase itself

What to do with CLAUDE.md:

1. Write minimal instructions
2. Focus on product, not tech
3. Add repetitive mistakes
4. Update regularly
5. Avoid “write good code” instructions (use linters)

Better Context > Bigger Context

- 200k (1M) tokens context window
- Better: 50k **important** context

Why:

- Model hallucinates less
- You spend less \$\$

Context — use responsibly

> /context

Context Usage

🗄️ 🗄️ 🗄️ 🗄️ 🗄️	claude-opus-4-5-20251101 · 30k/200k tokens (15%)
🗄️ 🗄️ 🗄️ 🗄️ 🗄️	🗄️ System prompt: 3.1k tokens (1.5%)
🗄️ 🗄️ 🗄️ 🗄️ 🗄️	🗄️ System tools: 16.6k tokens (8.3%)
🗄️ 🗄️ 🗄️ 🗄️ 🗄️	🗄️ MCP tools: 5.1k tokens (2.6%)
🗄️ 🗄️ 🗄️ 🗄️ 🗄️	🗄️ Memory files: 1.7k tokens (0.8%)
🗄️ 🗄️ 🗄️ 🗄️ 🗄️	🗄️ Skills: 1.4k tokens (0.7%)
🗄️ 🗄️ 🗄️ 🗄️ 🗄️	🗄️ Messages: 1.9k tokens (0.9%)
🗄️ 🗄️ 🗄️ 🗄️ 🗄️	🗄️ Free space: 125k (62.6%)
🗄️ 🗄️ 🗄️ 🗄️ 🗄️	🗄️ Autocompact buffer: 45.0k tokens (22.5%)

Context/sessions rules and commands

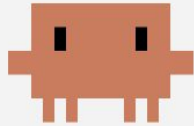
- New task — clear context with **/clear** command
- Already big chat — use **/compact** to summarize and continue
 - Sometime can lose important details
 - Sometimes better to use /clear
 - Can be autocompacted, but you don't have to wait
- Think you made a mistake in a previous prompt — **/rewind** command (or **Esc+Esc**)
 - works only with code modified by Claude Code
- Need to resume previous session — **/resume** command

Add Images to Context

Drag and drop or ctrl+v

Use-cases:

- Screenshots of UI
- Schemas



Claude Code v2.1.7

Opus 4.5 · Claude Max

~/Development/task-checker/task-checker-web

[Image #1] (↑ to select)

> **U**ry "edit submit/route.ts to..."

Tag files and folders with @

@file — automatically included into the context

@folder — include list of files from folder into the context

Otherwise agent has to find files on every session

Alternative: drag and drop files/folders

```
› add button to @app/page.tsx that will call api from @app/api/ ■
```

Include links in your prompts and ask to fetch

- Documentation
- Tutorials
- Public GitHub repositories

```
› use fetch tool to read https://docs.tokenfactory.nebius.com/quickstart  
and create a plan to integrate deepseek-ai/DeepSeek-R1-0528 model into  
our chat-bot
```

- **Fetch**(https://docs.tokenfactory.nebius.com/quickstart)
└ Received **324KB** (200 OK)

Ask agent to do web search for your

Use-cases:

- Find a tutorial
- Find a bugfix

```
› use web search and find tutorial on how to integrate trigger.dev  
background functions with next.js
```

```
Web Search("trigger.dev background functions Next.js integration tutorial  
2025")
```

```
| Found 10 results for "trigger.dev background functions Next.js integration  
tutorial 2025"
```

Bash tool (warning: dangerous)

Can execute commands for you

Use-cases:

- Git, gh
- Files manipulation
- Running tests and builds

btw, it is used for reading files (tail, head)

> revert the last commit

- **Bash**(git revert HEAD --no-edit)
└ Running...

Bash command

```
git revert HEAD --no-edit  
Revert the last commit
```

Do you want to proceed?

- > 1. Yes
2. Yes, and don't ask again for **git revert** commands in
/Users/denis/Development/cursor2/interviewprep
3. No

Esc to cancel · Tab to add additional instructions

Bash Tool Examples

in one of previous commits we removed a number of images from @public/ related to blog, restore them

I have @file.pdf made of a lot of scans, size ~100mb. I need to compress this pdf to <5mb. I already tried standard pdf compression methods. Create a script in python that will extract all the images from pdf, compress them separately and create a new pdf based on them. After that, run the script and check the output pdf size. If it is still too big, change the script and run it again. Repeat this process until pdf will be less than 5 mb. Don't remove initial pdf.

Plan mode (shift+tab)

- Use when creating more than one-purpose function
- It asks **clarifying questions**, explores codebase and creates a plan
 - Ask Claude to ask you clarifying questions(!)
- Iterate on the plan or review it in a separate session

Planning & Interaction

- `TodoWrite` – Manage task lists
- `AskUserQuestion` – Ask clarifying questions
- `EnterPlanMode` / `ExitPlanMode` – Plan implementations before coding

Claude Code Models

Available models:

- Claude Haiku
 - only for exploration
- Claude Sonnet
- Claude Opus

change model: /model

all 200k context



Claude Opus



Claude Sonnet



Claude Haiku

Claude Code Permissions

Default: read-only

Scope: all subfolders of a working directory

Tool type	Example	Approval required	"Yes, don't ask again" behavior
Read-only	File reads, Grep	No	N/A
Bash commands	Shell execution	Yes	Permanently per project directory and command
File modification	Edit/write files	Yes	Until session end

Brave enough?

claude --dangerously-skip-permissions

Skills & Subagents

Advanced Claude Code usage

Custom Skills

- On-demand knowledge
- Describe processes
- Can include other files, images, scripts
- Supported by majority of coding agents

my-skill/

├── SKILL.md	# Required: instructions + metadata
├── scripts/	# Optional: executable code
├── references/	# Optional: documentation
└── assets/	# Optional: templates, resources



Two types of custom skills

1. **Loaded by model** when needed

- a. works as **additional knowledge**
- b. example: custom library guidelines
- c. example: architecture description
- d. example: brand guidelines

2. **Custom commands**

- a. works as **pre-saved prompts**
- b. example: /techdebt — *“review uncommitted changes and make sure no duplicate code was introduced...”*
- c. example: /fixbuild *“run build command, make sure no build errors, fix if any”*
- d. example: /commit *“Generate commit message with Conventional Commits specification and commit changes”*
- e. add *disable-model-invocation: true* to SKILL.md

Code Review Skill

name: code-reviewer

description: Use this skill to review code. It supports both local changes (staged or working tree) and remote Pull Requests (by ID or URL). It focuses on correctness, maintainability, and adherence to project standards.

✓  code-reviewer

 SKILL.md

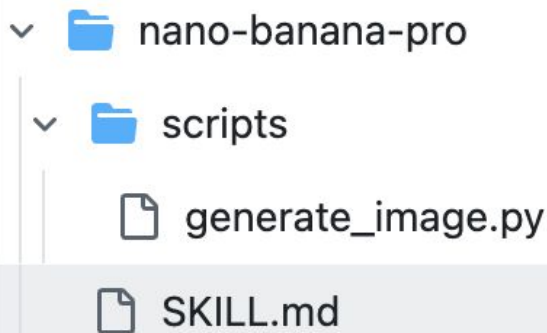
Nano Banana

name: nano-banana-pro

description: Generate or edit images via Gemini 3 Pro Image (Nano Banana Pro).

homepage: <https://ai.google.dev/>

metadata: {...}



PDF skill

name: pdf

description: Use this skill whenever the user wants to do anything with PDF files. This includes reading or extracting text/tables from PDFs, combining or merging multiple PDFs into one, splitting PDFs apart, rotating pages, adding watermarks, creating new PDFs, filling PDF forms, encrypting/decrypting PDFs, extracting images, and OCR on scanned PDFs to make them searchable. If the user mentions a .pdf file or asks to produce one, use this skill.

- ▼  pdf
- ▼  scripts
 -  check_bounding_boxes.py
 -  check_fillable_fields.py
 -  convert_pdf_to_images.py
 -  create_validation_image.py
 -  extract_form_field_info.py
 -  extract_form_structure.py
 -  fill_fillable_fields.py
 -  fill_pdf_form_with_annotati...
-  LICENSE.txt
-  SKILL.md
-  forms.md
-  reference.md

Background tasks library skill

name: trigger-tasks

description: Build AI agents, workflows and durable background tasks with Trigger.dev. Use when creating tasks, triggering jobs, handling retries, scheduling cron jobs, or implementing queues and concurrency control.

- ▼  trigger-tasks
 - ▼  references
 -  advanced-tasks.md
 -  basic-tasks.md
 -  scheduled-tasks.md
 -  SKILL.md

Skills best practices

- Use for knowledge that you need from time to time and repetitive tasks
- Mention skills in prompts explicitly: “use code-review skill to review uncommitted changes”
- Create /command skills for automations
- Store them in .agents/skills and create symlinks to .claude, .codex etc...
 - (ask Claude Code to create symlink)

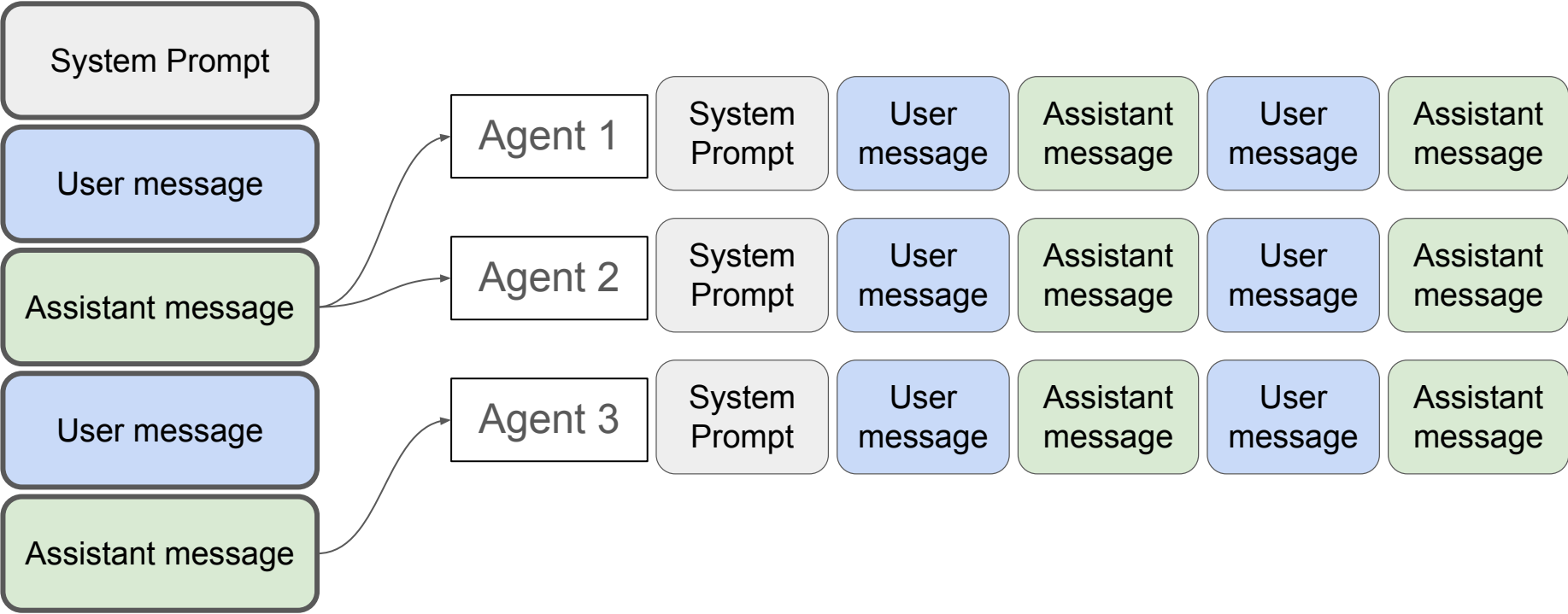
Subagents

Building a C compiler with a team of parallel Claudes

To test this system, we pointed it at an ambitious goal: building a web browser from scratch. The agents ran for close to a week, writing over 1 million lines of code across 1,000 files. You can explore the [source code on GitHub](#).

Subagents

Claude Code



Subagents: Task → Result

Assistant message

tool: call subagent X with prompt="..."

User message

result...

Subagents in Claude Code

Subagents work in their **own context, execute tasks and return results**

Existing subagents:

- General purpose agent — the same as main Claude
- Explore agent — searches and analyzes specific things in the codebase
- Plan agent — create a plan in Plan Mode

Use Explore Agent in Parallel

Uses **haiku model** (cheap)

prompt:

run multiple explore subagents in parallel to find out how X, Y, and Z is implemented

How to work with subagents

1. Brainstorm your task with Claude Code
2. prompt: “create a **detailed plan** for the implementation of X, save it in **md file in @plans folder**, include todos for every task”
3. prompt (new context): “implement @my_new_plan.md **step by step**. For every task, spin up **general purpose subagent**, wait for it to finish. After that, **check if the implementation is correct and mark task as done**. Then continue with the next task”.

Creating Custom Subagents

/agents command — generate automatically

› /agents

Create new agent

Choose location

- › 1. Project (.claude/agents/)
- 2. Personal (~/.claude/agents/)

› /agents

Create new agent

Describe what this agent should do and when it should be used (be comprehensive for best results)

e.g., Help me write unit tests for my code...

Creating Custom Subagents

Examples:

- math-articles-reviewer
- math-articles-critic

prompt:

use math-articles-reviewer agent to read the article @file.pdf let it return the review, then spin up math-articles-critic agent to find weaknesses of the article. summarize the results

Review Code With Subagents

prompt:

review local changes using general purpose agent: find all weak spots, code duplicated, architecture problems, security problems

don't fix it yet but provide me with a plan

Skills vs Agents

Skills

- Same context
- Custom prompts
- Can run code

Use: tasks that you repeat

Agents

- Separate context
- Custom prompts
- Can run code

Use: delegating autonomous work

Claude Code Pipeline

2026

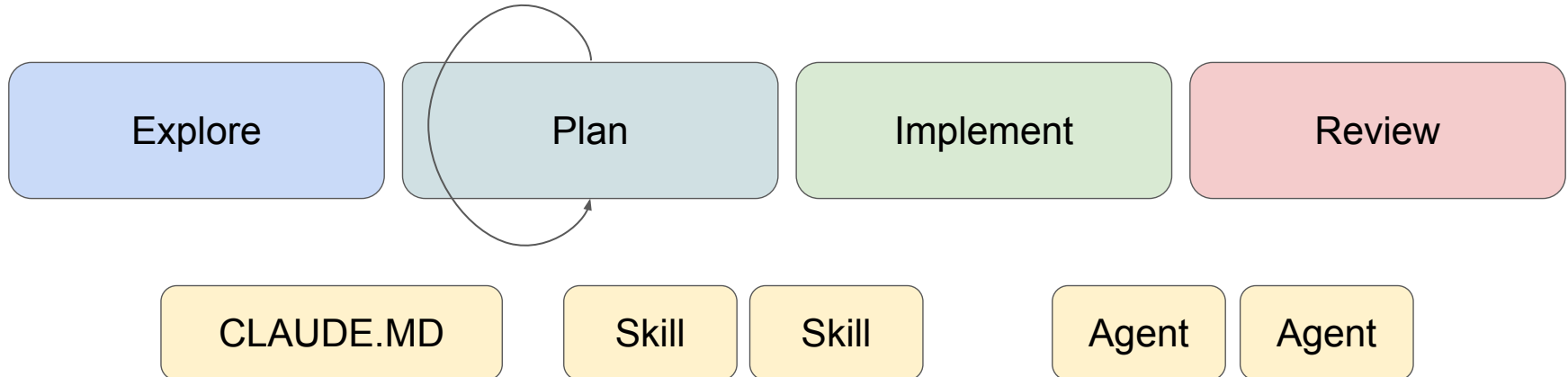


Claude Code is not perfect

- It doesn't follow your architecture
- It duplicates code
- It can make decisions and don't tell you about them
- It will not use existing libraries but create solutions itself
- It will hallucinate and not tell you about that

Your perfect AI coding pipeline

- Small/medium-sized features
- Review code in separate agent session (ask it to review uncommitted changes)
- Create [CLAUDE.md](#) and skills
- Use subagents for complex pipelines

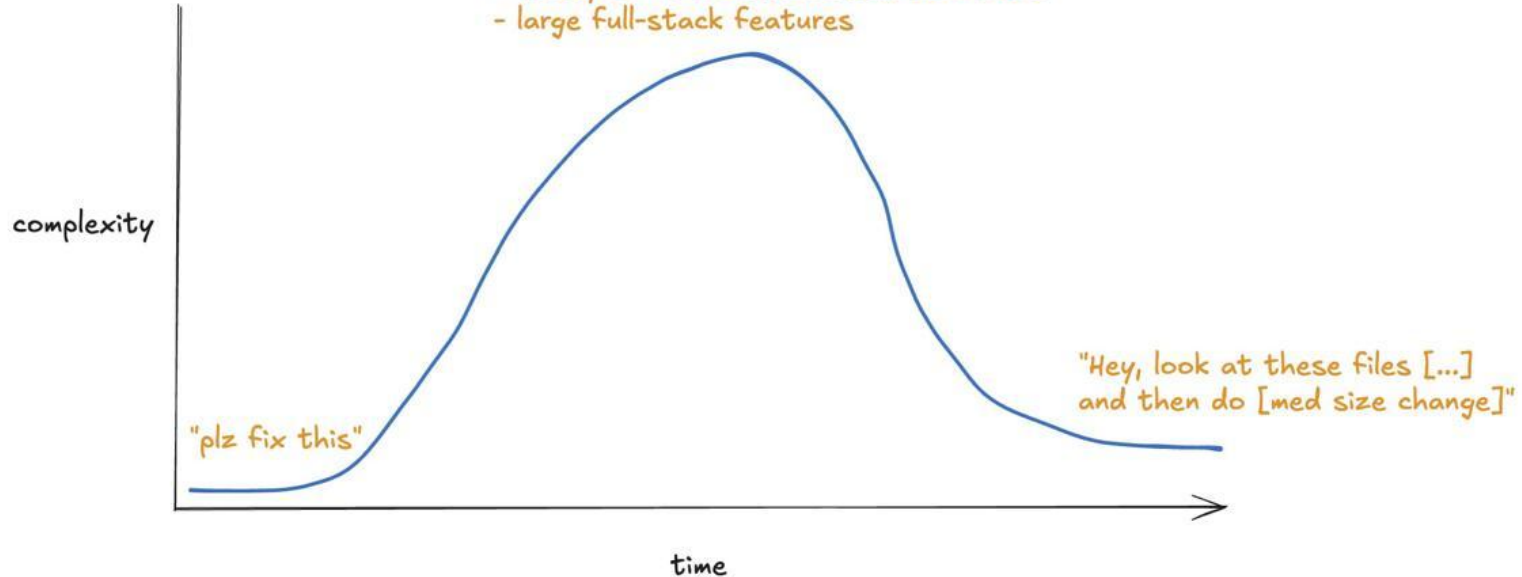


Great Claude Code use-cases

- code explanation and file search
 - ask to explore codebase in parallel
 - *run multiple explore subagents in parallel to find out how X, Y, and Z is implemented*
- internal and/or personal tools
 - now you have many junior devs not limited to your skills
 - *plan the dashboard visualizing graph we receive as output from @file using javascript*
- analyze logs
 - it can read files => it can analyze logs
 - it can run scripts and analyze output logs
 - *run @[main.py](#), check the error message, add debug logging and help me find the issue*
- write docs
 - you can set up automation for that
 - *based on your explanation create a comprehensive md file in @docs folder*

Curve of Agentic Programming

- 8 agents at same time
- complex orchestration with multi checkouts
- chaining agents together
- custom subagent workflows
- library of 18 different slash commands
- large full-stack features



Thank you for attention!

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